සියලුම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved

දකුණු පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව
தென் மாகாணக் கல்வித் திணைக்களம்
Department of Education, Southern Province

13 ශ්‍රේණිය අවසාන වාර පරීක්ෂණය - 2025 (ඔක්තෝබර්)
தரம் 13 ஆண்டிறுதிப் பரீட்சை - 2025 (ஒக்டோபர்) / Grade 13 Final Term Test - 2025 (October)

Information & Communication Technology I	20	E	I	කාලය நேரம் Time	02 Hours

නම பெயர் Name ..... විභාග අංකය சுட்டிலக்கம் Index No. .....

Should be considered:

- **Answer all the questions**
- **Write down your name and index number in the space provided in the answer sheet.**
- **For each question from 1 to 50, select the correct or most appropriate answer from the given answers of (1), (2), (3), (4) or (5).**

1. Which of the following is true regarding the data life cycle?
   A – It consists of three main steps.
   B – The data management steps also include data processing and data storage.
   C – The validity of unused data varies depending on the system in which data is used.
   (1) A only (2) B only (3) A and C only (4) B and C only (5) All A, B and C

2. The true statement regarding utility software is,
   (1) All utility software is installed on a computer along with application software
   (2) This software is installed on a computer along with the operating system or after the operating system is installed
   (3) Files are stored permanently on the hard disk through disk defragmentation
   (4) These are essential for the development of computer programmes
   (5) Installing this software reduces the performance of a computer

3. Which of the following statements are true?
   A – The world's first automatic computer was the Mark 1.
   B – The world's first electronic digital computer was the EDVAC.
   C – The world's first digital computer was the ENIAC, which operated on stored program concept.
   D – The Analytical Engine is considered the world's first computer model.
   (1) A and C only (2) A and B only (3) A and D only (4) B and D only (5) C and D only

4. Which of the following statements is incorrect regarding the bus lines used to exchange data in the form of electronic signals between computer devices?
   (1) The control signals issued by the control unit are transmitted via the control bus.
   (2) The address bus facilitates data transmission between two computer devices.
   (3) To write or read data from computer memory, the appropriate location is directed by the address bus.
   (4) The address bus is used to find a location data location in computer memory.
   (5) Data in computer memory is moved from one location to another by the data bus.

5. Consider the following statements regarding the solution of the operation $(47_{10}) + (-137_{10})$ using the eight-bit two's complement method.
   A – The two's complement value of the eight bit of 47_{10} is 00101111
   B – The two's complement value of the eight bit of 47_{10} is 11010001
   C – The above operation cannot be solved using the two's complement method of eight bits

D – When the above operation is solved using the two's complement of eight bits method, the answer is 10100110

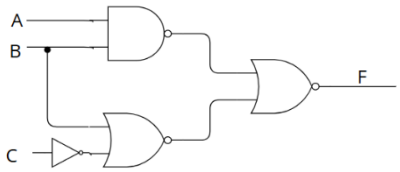
Which of the above statements are true?

(1) A and C only (2) A and D only (3) B and C only (4) B and D only (5) None of the above

6. The code for representing a character on a computer is 10101101. Which encoding method is used to encode this character?

(1) ASCII coding method (2) Unicode method (3) BCD coding method (4) EBCDIC coding method
(5) BCNF coding method

7. Which expression represents the output F of the following logical circuit?



(1) $AB\bar{C}$ (2) AB (3) $A\bar{C}$ (4) $(AB) + (B + \bar{C})$ (5) $AB.(B + C)$

8. What is the answer when the Boolean expression $(A + \bar{C}).A + \overline{(A.B)}$ is simplified?

(1) B (2) A (3) C (4) 0 (5) 1

9. When the Boolean expression $F = A.B + B.C + A$ is plotted on a K-map, which answer contains the correct K-map?

(1)

C \ AB	00	01	11	10
0	0	1	1	1
1	0	0	1	1

(2)

C \ AB	00	01	11	10
0	0	0	1	1
1	0	1	1	1

(3)

C \ AB	00	01	11	10
0	1	0	0	0
1	1	1	0	0

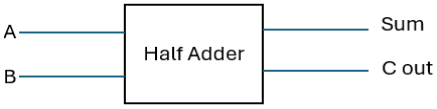
(4)

C \ AB	00	01	11	10
0	1	1	0	0
1	1	0	0	0

(5)

C \ AB	00	01	11	10
0	0	1	1	0
1	0	1	1	1

10. Which answer correctly contains the **Sum** output and the **Cout** output when input A is 1 and input B is 0?



(1) 1, 1 (2) 1, 0 (3) 0, 1 (4) 0, 0 (5) None of the above

11. A logical memory address consists of a page number and an offset. A physical memory address consists of a frame number and an offset. Why is the offset the same in both addresses?

(1) The offset is used to identify the specific process that generated the address
(2) The offset refers to a specific byte of data, and the location of that byte within a page or frame does not change during translation
(3) The offset is a constant value that helps the MMU identify the correct page table
(4) The offset is a remainder of older memory management systems and does not serve any essential function in modern systems
(5) The offset is used to determine whether a page fault has occurred

12. The size of one cluster/block on a hard disk is 16KB. The table shown here shows a portion of the FAT table (File Allocation Table) on the hard disk.

Block No	Value
100	102
101	-1
102	104
103	101
104	103

Note:
1. The last category of a file is indicated by -1.
2. The first category of the file data.txt is 100.

Which of the following shows the total amount of space allocated on the hard disk for the file data.txt?

(1) 48 KB (2) 64 KB (3) 164 KB (4) 32KB (5) 80 KB

13. What is the reason for a process to transit from the Blocked state to the Ready state?
(1) It requests an I/O (Input/Output) operation
(2) The process is newly created
(3) The I/O operation, it requested is completed
(4) Its time slice ends
(5) A process terminates

14. When is the current state of a process updated?
(1) Only when a process is started
(2) Once every second
(3) When the user enters new data into the process
(4) Every time a process changes its state
(5) When the CPU is idle

15. Which class does the IP address 229.0.10.153 belong to?

(1) E (2) D (3) C (4) B (5) A

16. What is the number of network bits and host bits respectively that should be used to create seven subnets from a computer network with the network address 192.168.10.0?

(1) 1 and 7 (2) 2 and 6 (3) 3 and 5 (4) 4 and 4 (5) 3 and 6

17. Which answer shows the network address and subnet mask of the network respectively to the computer with the IP address 192.168.5.7/25 belongs?

(1) 192.168.5.192 and 255.255.255.0 (2) 192.168.5.0 and 255.255.255.0
(3) 192.168.5.0 and 255.255.255.192 (4) 192.168.5.64 and 255.255.255.0
(5) 192.168.5.128 and 255.255.255.192

18. Which of the following server performs tasks such as providing internet services to user computers on the local area network and imposing restrictions on the websites accessed by users on that network?

(1) Domain name server (DNS server) (2) Proxy server (3) Web server
(4) Mail server (5) Application server

19. Consider the following statements regarding signal encoding methods.

A – In encoding, data bits are mapped to data packets.
B - Representing a data signal by a positive voltage representing a binary value of 1 and a negative voltage representing a binary value of 0 is called “Non return to zero (NRZ-L)” method.
C - When a data signal is represented by a non-return to zero inversion (NRZ-i) method, a physical transmission occurs when the binary value is 0, and no physical transmission occurs when the binary value is 1.
D - In Manchester encoding, a transition occurs in the middle of each bit cycle.

Which of the above statements are true?

(1) A and B only (2) B and C only (3) C and D only (4) A and D only (5) B and D only

20. Which of the following is false about the ALOHA protocol?

(1) The first public demonstration of wireless data packets implemented the operations of the ALOHA protocol
(2) Slotted ALOHA is not an improvement on the ALOHA protocol
(3) Slotted ALOHA introduced discrete time slots for data transmission
(4) Wireless data packets can only be transmitted to a service point at the beginning of a time slot
(5) Slotted ALOHA increased the maximum achievable time for wireless data packet transmission

21. When carrying out cultivation activities in a protected house, the soil condition is studied to apply fertilizer to the plants in the protected house and the necessary fertilizer is applied to the plants.
What is the most suitable information system for that?

(1) Transaction Support System (TPS)
(2) Knowledge Management System (KMS)
(3) Management Information System (MIS)
(4) Decision Support System (DSS)
(5) Content Management System (CMS)

22. Select the correct answer regarding the System Development Life Cycle (SDLC).

(1) During the information system review, a preliminary investigation of the existing system is carried out.
(2) Data flow diagrams are created during the system analysis phase.
(3) A critical investigation is carried out to determine how well the proposed system is suitable for the individual or organization through a feasibility study.
(4) During the system development phase, coding is carried out using an appropriate computer language selected by the programmers.
(5) During the system implementation phase, changes affecting the system are checked and updated regularly.

23. Which of the following is a false statement about the Agile Model?

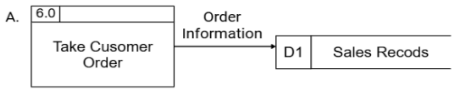
(1) The Agile Model has an iterative approach, and after each iteration, a part of the software is released to the users.
(2) The Agile Model can be used to design systems with constantly changing requirements.
(3) The Agile Model cannot be used to design systems with fixed requirements.
(4) There should be constant discussions between the client and the system design organization during the system design process.
(5) By designing systems using the Agile Model, the client can be provided with a higher quality system in a much shorter time.

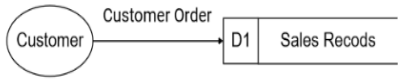
24. Consider an information system that can access 1000 patients at a time, with a patient being able to access the system within two seconds at any time of the day.

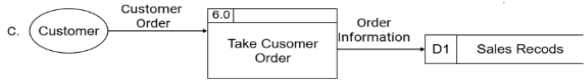
A. - 1000 users should be able to access the system at the same time
B. - The system should be operational 24/7 at any time of the day.
C. - Users should be able to access the system in two seconds or less.
Which statement/statements can be considered as a functional requirement related to the above system?

(1) A only (2) A and B only (3) A and C only (4) B and C only (5) All A, B and C

25. Which diagram represents a correct data flow?

A.


B.


C.


(1) A only (2) A and B only (3) A and C only (4) B and C only (5) All A, B and C

26. Select the answer that contains a false statement about testing methods.
(1) White box testing can be done by programmers and requires knowledge of programming.
(2) Unit testing can be done to minimize errors that may occur when newly added features and existing features are changed.
(3) System testing does not test both functional and non-functional requirements.
(4) Black box testing is a test performed from the perspective of system users.
(5) Integration testing can be done to minimize errors that occur when several independent software are combined.

27. The relationship between "Employee" and "Project" is often Many-to-Many. What happens when this relationship is implemented in a database?

(1) By inserting information about "Project" into the "Employee" table
(2) By inserting information about "Employee" into the "Project" table
(3) By creating an intermediate table that connects employees and projects
(4) By merging the "Employee" and "Project" tables
(5) This cannot be implemented in a database

28. What is a weak entity?
(1) An entity that can exist on its own, but does not have complete meaning without a relationship with another entity
(2) An entity that does not exist on its own, and is completely dependent on another strong entity
(3) An entity that has a unique primary key
(4) An entity that cannot have more than one attribute
(5) An entity that has no attributes

29. The following is a relation R with the set of attributes R= {A, B, C, D, E} and its functional dependencies.
A -> B, C
C -> D
B, D -> E
The primary key of R is A.
What is the normal form of this relation R?
(1) It is in first normal form (1NF) and not in second normal form (2NF)
(2) It is in second normal form (2NF) and not in third normal form (3NF)
(3) It is in second normal form (2NF) and also in third normal form (3NF)
(4) It is only in third normal form (3NF)
(5) It is not in any normal form

30. What is the correct SQL statement to change the salary value of the employee with emp_id = 101 to 60000 in the Employee table?
(1) Update Employee set salary = 60000 where emp_id = 101;
(2) Alter table Employee set salary = 60000 where emp_id = 101;
(3) Modify Employee set salary = 60000 where emp_id = 101;
(4) Insert into Employee set salary = 60000 where emp_id = 101;
(5) Update salary = 60000 from Employee where emp_id = 101;

31. Which of the following statements is not correct as a characteristic of a relational database?

(1) The value of a primary key must always be unique
(2) The value of a foreign key must match the domain of a primary key
(3) Null values cannot be placed in every field of a table.
(4) A composite key is a key formed from two or more fields of a table
(5) A foreign key refers to the primary key of another table.

32. What are the main risks of the Flat File data model?

(1) Ability to process relational data
(2) Ensuring that data remains accurate without changing
(3) Data repetition
(4) Acting as a structured database
(5) Data consistency

33. In Python programming, when a=5, b=3, the answer to (a ^ b) is,

(1) 001_2 (2) 1_{10} (3) 6_{10} (4) 110_{10} (5) 7_{10}

34. The output of this Python code will be,

```
Z=10
While (Z=2):
Print(end=",")
Z - =3
If (Z  = = 4):
    break
else:
    print(z,end=",")
```

(1) 10, 0, 8, 0, 6 (2) 10,8,6,0 (3) 0,10,8,6,0

(4) 0 (5) Error

35. The output of the following Python function is,

```
def additem(listparam):
        listparam+=[1]
mylist=[1,2,3,4]
addItem[mylist]
print(len(mylist))
```

(1) 5 (2) 2 (3) 3 (4) 8 (5) 1

36. What is the output of the following python code?

```
X = 12
def  F1(a,b=x):
      print(a,b)
X=15
F1(4)
```

(1) Error (2) 12 4 (3) 4 12 (4) 4 15 (5) 4 4

37. What is the output of the following Python program?

```
j = 10
y = 4
If  j<y and j!=4:
        J - = y
        print(j)
else ;
        y *=j
        print +(y)
```

(1) 6 (2) 40 (3) 10 (4) 14 (5) 2

38. The purpose of using triple quotes for a Python string is to,

(1) Use only single or double quotes for strings
(2) Use to indicate multi-line strings
(3) Use to indicate empty strings in the string
(4) Use to indicate multi-line comments
(5) None of the above

39. Which of the following answers represents the content of the file 2.txt, which is created using the file 1.txt through the following Python program?

Ruwan, 20, 50
Ramesh, 0, 5

```
f1=open ("1. txt","r")
f2=open ("2. txt","w")
for line in f1:
        items=line.strip ( ).split (",")
        tot=int (items[1])+int (items[2])
        print (items[0], items[1], items[2], tot, file=f2)
f1.close ( )
f2.close ( )
```

(1)	(2)	(3)	(4)	(5)
Ruwan 20 50 70	20 50 70	70	70 5 20	Ruwan 70
Ramesh 0 5 5	0 5 5	5		Ramesh 5

40. The output for the following given code will be,

```
X = Y = [5, 3]
X.append (2)
Y.append  (X)
print (x,y)
```

(1) [5, 3, 2] [5, 3, 5, 3, 2]
(2) [5, 3, 2] [5, 3, 2]
(3) [5, 3 2] [5, 3, 5, 3, 2]
(4) [5, 3, 2] [5 3 2, 5 3 2]
(5) [5 3 2] [5 3 2, 5 3 2]

41. Consider the following statements about an HTML element.

A. - It contributes to the creation of a single piece of content on a web page
B. - An element consists of a start tag, an end tag, and content
C. - <h1>Sri Lanka</h1> is an example of the correct use of an HTML element

Among the above statements, the correct statement(s) is/are,

(1) A only (2) A and B only (3)B and C only (4) A and C only (5) All of A, B and C

42. The correct tags and syntax to use to include an external CSS file is,

(1) <link href="style.css" rel="stylesheet">
(2) <style src="style.css">
(3) <css link="style.css">
(4) <link style="style.css">
(5) <style link="style.css" rel="stylesheet">

43. The correct answer for how to correctly apply an internal style sheet in an HTML document is,

(1) <style>p{ color: blue; } </style>
(2) <link>p{ color: blue; } </link>
(3) <script>p{ color: blue; } </script>
(4) <style href="styles.css">
(4) <p style="color: blue" >

44. The output of the following PHP code could be,

```
<?php
$name = "Hello";
$number = 1001;
echo $name . " " . $number;
?>
```

(1) Hello1001 (2) Hello" "1001 (3) Hello 1001 (4) Hello." ".1001 (5) $name." ".$number

45. The output of the following HTML code could be,

```
<html>
<table border="1">
 <tr>
        <td colspan="2"> Lanuages: </td>
 </tr>
 <tr>
        <td>
                <ul>
                 <li>Python</li>
                 <li>Java</li>
                </ul>
        </td>
         <td> High-Level </td>
 </tr>
</table>
</html>
```

(1)

Lanuages:	
• Python • Java	High-Level

(2)

Lanuages:	
• Python • Java	High-Level

(3)

Lanuages:	
1. Python 2. Java	High-Level

(4)

Lanuages:	
Python Java	High-Level

(5)

Lanuages:	
• Python • Java	High-Level

46. Consider the following statements about the basic components that are generally included in an embedded system,

A - Must contain a microcontroller
B - May or may not have an operating system and sensors
C - Must provide an interface through a web browser

The correct statement/s from the above statements is/ are,

(1) A only (2) B only (3) C only (4) A and B only (5) B and C only

47. Which of the following is not a major challenge of the large volume of data for the Internet of Things?

(1) Data security and privacy
(2) Data processing and real-time analysis
(3) Data storage and management
(4) Non-conformant data formats
(5) Power consumption of sensors

48. A characteristic of a virtual store could be,

(1) Having only a physical store location
(2) Being open to customers worldwide
(3) Requirement of more service staff
(4) Limited hours of store opening
(5) Products are displayed but not available for purchase

49. An example of a B2C (Business to Consumer) e-commerce transaction would be,

(1) A manufacturer orders raw materials from a supplier
(2) A person buys a pair of shoes online
(3) Exchange of technical data between two companies
(4) A customer sells used goods through a website
(5) A customer makes a payment using GovPay

50. Which of the following characteristic is not possessed by a software agent

(1) Autonomy
(2) Reactivity
(3) Manual control
(4) Pro-activeness
(5) Social Ability